Q1. Joe bought a potted plant. He kept it well watered but some of the leaves turned yellow.



Joe thought that the plant did **not** have enough light for photosynthesis. He moved the plant closer to the window but more leaves turned yellow.

(a) He then thought that the plant did **not** have enough minerals.

The table below gives information about minerals.

mineral	why the mineral is needed
magnesium	to make chlorophyll
nitrogen	to make protein
phosphorus	to grow and transfer energy
potassium	to make fruit

(i) Joe's plant did **not** have enough of one of the minerals in the table.
Use the information in the table to suggest which mineral this was.

.....

1 mark

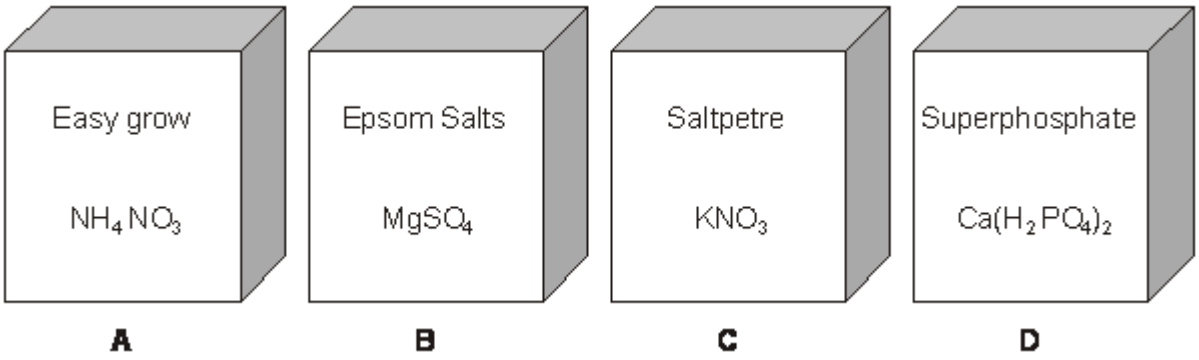
(ii) A plant growing in a pot is more likely to be affected by a shortage of minerals than a plant growing in a garden.
Give the reason for this.

.....

.....

1 mark

(b) Joe bought some fertiliser for his plant.
The names and formulae of four different fertilisers are shown below.



(i) Give the letter of **one** box of fertiliser, A, B, C or D, that would provide each of the minerals in the table below.
Write the letters in the table.

mineral	**letter of fertiliser**
magnesium	
nitrogen	
phosphorus	
potassium	

3 marks

(ii) Easy Grow is ammonium nitrate, NH_4NO_3.

How many different elements are present in ammonium nitrate?

.....

1 mark

(iii) How many atoms are present in the formula of ammonium nitrate?

.....

1 mark
maximum 7 marks

Q1.

(a) (i) magnesium

1 (L6)

(ii) any **one** from

- a potted plant has a limited amount of soil **or** minerals
  accept the converse
  *answers must indicate that the soil available to the potted plant is limited **or** may not be replenished*
- less chance of minerals being returned to the soil by dying **or** decaying plants **or** by microbes **or** fungi **or** bacteria **or** worms
  'the garden has more minerals' is insufficient
  'lack of sunlight' is insufficient
- no room for roots to extend and find more minerals
  accept 'no room for roots to grow'

1 (L6)

(b) (i)

mineral	letter of fertiliser
magnesium	B
nitrogen	A **or** C
phosphorus	D
potassium	C

*accept 'Epsom Salts' **or** '$MgSO_4$' **or** 'magnesium sulphate'*
*accept 'A and C **or** 'Easy Grow' **or** 'NH_4NO_3'*
***or** 'ammonium nitrate' **or** 'Saltpetre' **or** 'KNO_3'*
***or** 'potassium nitrate'*
*accept 'Superphosphate' **or** '$Ca(H_2PO_4)_2$'*
*accept 'Saltpetre' **or** 'KNO_3' **or** 'potassium nitrate'*
if all four answers are correct, award three marks
if three answers are correct, award two marks
*if one **or** two answers are correct, award one mark*

3 (L6)

(ii) 3

1 (L6)

(iii) 9

1 (L6)

[7]